

Autodesk® Scaleform®

Scaleform Best Practices Guide

Scaleform 4.0

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: 3.0
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1		1
1.1		3
1.2		3
2		4
2.1		4
2.2		4
2.3		5
2.3.1	vs	5
2.3.2		5
2.3.3		7
2.3.4		9
2.3.5		11
3	ActionScript	13
3.1	ActionsScript	13
3.1.1		16
3.1.2		16
3.1.3	/	16
3.2	ActionScript 3	17
3.2.1		17
3.3	Advance	18
3.4	onEnterFrame and Event.ENTER_FRAME	18
3.5	ActionScript 2	19
3.5.1	onClipEvent on	19
3.5.2	Var	20
3.5.3		21
3.5.4		22
3.5.5		22
4	HUD	24
4.1	SWF	24
4.2	SWF 1	25
4.3	1	26
4.4	1 ().....	26
4.5	Flash HUD	27
5		29
5.1	Flash	29
5.2		29
6		30

Autodesk® Scaleform® 4.0

Adobe Flash®

Flash

ActionScript™ (AS)

(HUD)

HUD

HUD

Scaleform 3.0

Flash

Web

Scaleform

Flash AS

(

PC

)

(

PC

)

(UI)

UI

Flash

FLA

Flash Studio CS5

XML
.FLA

1

Flash

AS

AS

Flash

FLA

Flash

:

• Flash

AS

:

◦

- FLA AS :
- AS

- :
-

- Flash AS :
- Flash

Flash

Flash AS

Flash :

-
-

ActionScript

1

Flash

Flash

一般的な使用法

- 1
- UI (Flash [])

Scaleform 3.3
Scaleform Common Lightweight Interface Kit (CLIK™)

セットアップの検討事項

-
- - Flash
 - Flash
 - AS
 - AS2
 - AS3
 - MMORPG
 - HUD
 - SWF
 - API
 - C++ API (Gfx::Movie
 - Scaleform 3.3
 - Flash
 - CPU
 - CD
 - DVD
- - SWF
- - UI
 - :
 - HUD : 400K-1.5M
 - / : 800K-4M
 - 700K-1.5M ActionScript ():
 - : 2M-10M+

2.

3. AS2) and `visible=false` rather than `alpha=0` (in AS3) `_alpha=0` `_visible=false`(in "stop()")

アートワーク

Flash

Scaleform

(
()
)

•

•

○

(

Flash

•

Player

○

O

Flash

Scaleform

•

○

○

•

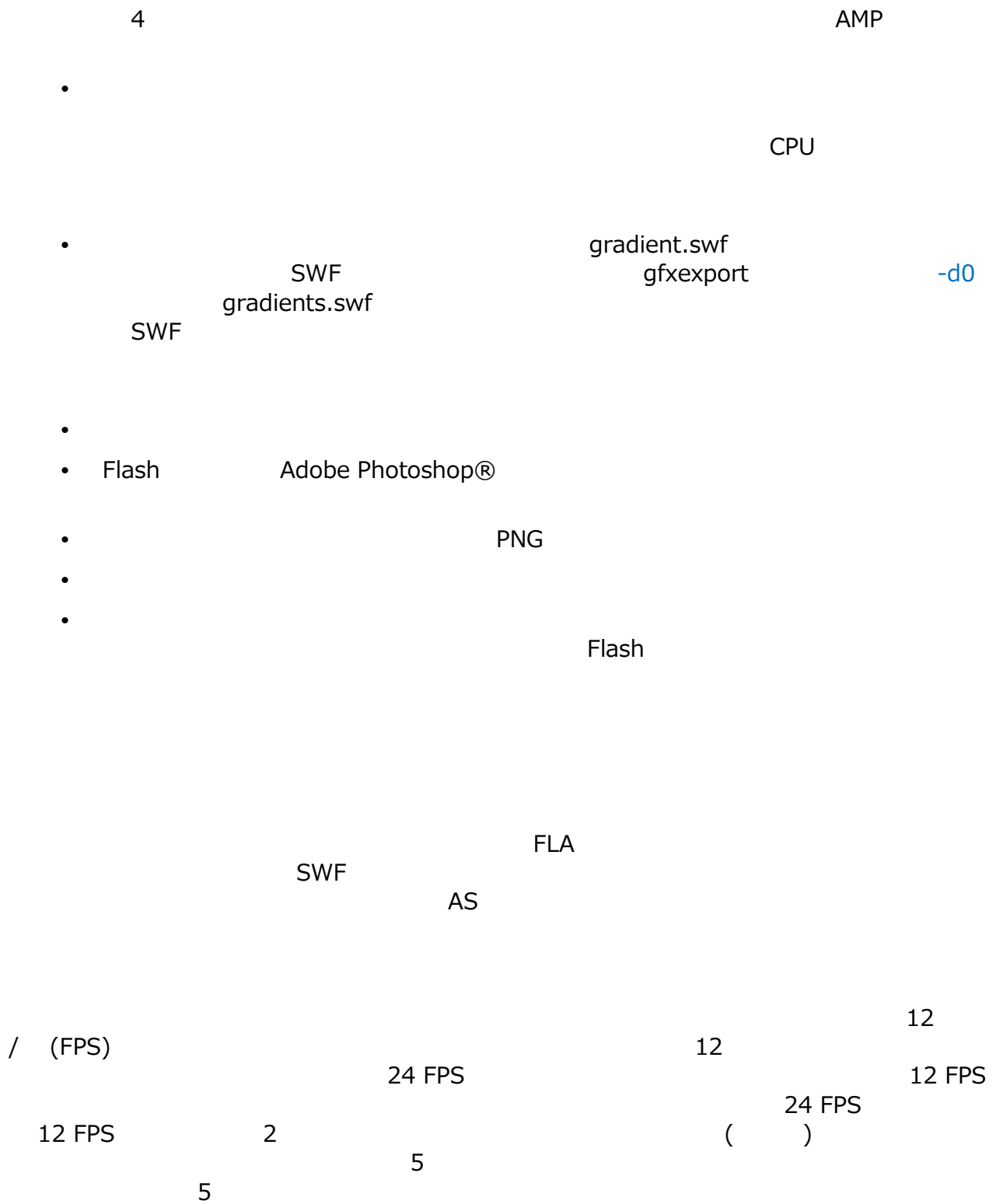
()

○

O

◦			2
•	Flash	API	1
•			
		Flash Studio	
	Scaleform	EdgeAA	
•			
•			
•	EdgeAA		1
	/	/	

Flash 10



onEnterFrame(in AS2) or Event.ENTER_FRAME (in AS3)

onEnterFrame/Event.ENTER_FRAME

setInterval

onEnterFrame/EVENT.ENTER_FRAME

setInterval

UI

30 40 FPS
CPU

Flash

:

•

•

/

/
_visible/visible = true/false

•

◦

1

◦

Flash

◦ Tween
Scaleform CLIK Tween (gfx.motion.Tween in AS2 or
scaleform.clik.motion.Tween in AS3)
Tween

Flash

•

60

Flash

Flash

60 FPS

•

•

•

(

)

•

-

-

CPU

(
()

CPU
)

4

/

-

()

-

gfxexport

(48)

SlotHeight

DP
)

(

DP

-

DP 1

-

1

30 FPS

-

("TextField.variable")

- "htmlText"

HTML

- gfxexport -fc -fcl -fcm

()

"Font and Text Configuration Overview"

- fontlib
("Font and Text Configuration Overview"
)
- TextField 1 1
1 DP
- 1 (1)
- ()
-
- DrawText API DrawText API
Flash Scaleform C++
Flash UI DrawText API C++ 1

ActionScript

AS

AS

一般的な *ActionScript* のガイドライン

AS

- SWF Flash 8
- AS

AS

[attachMovie](#)

FLA

- AS
-
-
- “if”

- [on\(\)](#) [onClipEvent\(\)](#) [onEnterFrame](#)
[onPress](#)

- AS

Flash AS

([onClipEvent](#) on)

([gotoAndPlay](#) play stop)

- , [gotoAndPlay/gotoAndStop](#) ,

- gotoAndPlay/gotoAndStop
 gotoAndPlay/gotoAndStop
 gotoAndPlay/gotoAndStop
- gotoAndPlay/gotoAndStop
 gotoAndPlay/gotoAndStop
- gotoAndPlay/gotoAndStop
 gotoAndPlay/gotoAndStop
 gotoAndPlay/gotoAndStop
- Gfx::Movie::SetVariableArray C++ C++ AS 1
 1 1
- 1 Scaleform
 Scaleform
- Object
 ◦ Object
- eval() 1
- myArr.length Array.length
 :
- ```
var fontArr:Array = TextField.getFontList();
var arrayLen:Number = fontArr.length;
for (var i:Number = 0; i < arrayLen; i++) {
 trace(fontArr[i]);
}
```

```

:
var fontArr:Array = TextField.getFontList();
for (var i:Number = 0; i < fontArr.length; i++) {
 trace(fontArr[i]);
}

```

- ( null )
- removeListener()
- - AS VM
  - com.xxx.yyy.aaa.bbb VM if AS
  - aaa bbb com xxx yyy if
  - com xxx xxx yyy
  - Flash SWF
  - c58923409876.functionName()
- - AS SWF SWF
- AS
- - SWF [ ] trace() [Flash] [Trace
  - ] trace
- 
- - 1 SWF AS ( : foo.bar.CustomClass)
  - SWF SWF
  - SWF

delete

:

delete foo.bar.CustomClass

- ( 100% )

- onEnterFrames

- ( ) Scaleform

- 
- 
- 
- 
- 

- 
- with

- 
- "var" "var"  
ActionScript

-

- 
- null
- SWF  
null
- AS getter setter

## ActionScript 3 の汎用ガイドライン

AS3

- 
- Object Object AS3
- Array Array Vector  
4294967295  
Array
- Vector Vector<\*>  
Vector<\*>  
Vector<String> Vector<YourFavoriteClass> Vector<int>  
Vector<\*>
- Object  
flash.utils.Dictionary
- GfX Flash
- AS2 AS3 AS3

## Advance

Advance

6 :

1. AS `onEnterFrame/Event.ENTER_FRAME`
2. `UI` `Invoke`
3. `(AS3)` `true` `stop()` `(_visible(AS2) or visible`  
`)` `Advance` `1 1`

4. `_global.noInvisibleAdvance(AS2)` or `scaleform.gfx.Extensions.noInvisibleAdvance`  
`(AS3)`

`Advance` `"true"`  
`Advance` `(`  
`)` `Flash`  
`Flash` `(`  
`)` `_global.gfxExtensions(AS2)` or  
`scaleform.gfx.Extensions.enabled (AS3)` `true` `Scaleform`

5. `Advance`
- 6.

## ***onEnterFrame and Event.ENTER\_FRAME***

`onEnterFrame(AS2)` or `Event.ENTER_FRAME (AS3)`

`onEnterFrame/Event.ENTER_FRAME`  
`setInterval` `setInterval` `setTimeout`  
`setInterval` :

- clearInterval
- setInterval    setTimeout    onEnterFrame/Event.ENTER\_FRAME  
onEnterFrame/Event.ENTER\_FRAME

onEnterFrame                      delete                      :

```
delete this.onEnterFrame;
delete mc.onEnterFrame;
```

onEnterFrame    null    undefined                      (    this.onEnterFrame = null;)  
onEnterFrame                      onEnterFrame  
Scaleform

## ***ActionScript 2 の最適化***

onClipEvent()    on()                      onEnterFrame    onPress  
:

- 
- onClipEvent    on  
this    \_global    arguments    super  
256

onLoad

onEnterFrame                      :                      onClipEvent(construct)

```
onClipEvent(construct)
{
 this.onLoad = function()
 {
 //
 }
}
```

onClipEvent(load)

var

AS

var

AS

2

:

```
var i = 1000;
countIt = function()
{
 num = 0;
 for(j=0; j<i; j++)
 {
 j++;
 num += Math.random();
 }
 displayNumber.text = num;
}
```

:

```
var i = 1000;
countIt = function()
{
 var num = 0;
 var ii = i;
 for(var j=0; j<ii; j++)
 {
 j++;
 num += Math.random();
 }
 displayNumber.text = num;
}
```



) (var

```
function foo(var obj:Object)
{
 for (var i = 0; i < obj.size; i++)
 {
 obj.value[i] = obj.num1 * obj.num2;
 }
}
```

```
function foo(var obj:Object)
{
 var sz = obj.size;
 var n1 = obj.num1;
 var n2 = obj.num1;
 for (var i = 0; i < sz; i++)
 {
 obj.value[i] = n1*n2;
 }
}
```

```
var floor = Math.floor
var ceil = Math.ceil
num = floor(x) - ceil(y);
```

```
var keyDown = Key.isDown;
var keyLeft = Key.LEFT;
if (keyDown(keyLeft))
{
 // ;
}
```

:

```
mc.ch1.hc3.djf3.jd9._x = 233;
mc.ch1.hc3.djf3._x = 455;
```

:

```
var djf3 = mc.ch1.hc3.djf3;
djf3._x = 455;

var jd9 = djf3.jd9;
jd9._x = 223;
```

C

:

```
this[_global.mynames[queue]][_global.slots[i]].gosplash.text =
_global.MyStrings[queue];
```

:

```
var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
var slot_i = _global.slots[i];
_splqueue[slot_i].gosplash.text = _splstring;
```

:

```
for(i=0; i<3; i++)
{
 this[_global.mynames[queue]][_global.slots[i]].gosplash.text =
 _global.MyStrings[queue];
 this[_global.mynames[queue]][_global.slots[i]].gosplash2.text =
 _global.MyStrings[queue];
 this[_global.mynames[queue]][_global.slots[i]].gosplash2.textColor =
```

```
0x000000;
}
```

:

```
var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
for (var i=0; i<3; i++)
{
 var slot_i = _global.slots[i];
 _splqueue[slot_i].gosplash.text = _splstring;
 _splqueue[slot_i].gosplash2.text = splstring;
 _splqueue[slot_i].gosplash2.textColor = 0x000000;
}
```

:

```
var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
for (var i=0; i<3; i++)
{
 var slot_i = _global.slots[i];
 var elem = _splqueue[slot_i];
 elem.gosplash.text = _splstring;
 var gspl2 = elem.gosplash2;
 gspl2.text = splstring;
 gspl2.textColor = 0x000000;
}
```

(HUD)

Scaleform HUD

HUD

4.1

HUD

HUD

HUD

C++

## 複数の *SWF* ムービー ビュー

HUD

80K

HUD

SWF

:

1. 別々のスレッド上で を呼び出す機能。

Flash

HUD

Advance

( )

AS

Flash

Advance

Flash

Advance

HUD

Advance

Advance

HUD

Advance

: Scaleform

GFx::Movie

Advance

|    |              |                    |       |
|----|--------------|--------------------|-------|
|    | Invoke       | Display<br>Advance | Input |
| 2. | キャッシングを利用する。 | HUD                |       |
|    |              |                    | HUD   |

## 複数の SWF を含む 1 つのムービー ビュー

|  |       |                           |   |
|--|-------|---------------------------|---|
|  | Flash | AS <code>loadMovie</code> | 1 |
|--|-------|---------------------------|---|

Display

|  |       |   |
|--|-------|---|
|  | Flash | 1 |
|--|-------|---|

1.

|  |                                                                |                                                 |
|--|----------------------------------------------------------------|-------------------------------------------------|
|  | <code>visible = false (AS3)</code>                             | <code>_visible = false(AS2) or</code>           |
|  | <code>scaleform.gfx.Extensions.noInvisibleAdvance (AS3)</code> | <code>_global.noInvisibleAdvance(AS2) or</code> |
|  |                                                                | <code>true</code> Advance                       |

( `_visible/visible = false` )

Advance  
Flash  
HUD / (

) HUD / (

Advance `scaleform.gfx.Extensions.enabled (AS3)` `true` `_global.gfxExtensions(AS2) or`  
Scaleform

2.

|   |                               |        |        |
|---|-------------------------------|--------|--------|
|   | <code>SetVariableArray</code> | 1      | Invoke |
| ( |                               | )      | 1      |
|   | <code>SetVariableArray</code> | (      |        |
| ) | 1                             | Invoke |        |

1

3. HUD onEnterFrame(AS2) or  
Event.ENTER\_FRAME (AS3)  
HUD onEnterFrame/Event.ENTER\_FRAME  
Advance  
onEnterFrame
4. AS  

tick  
AS Invoke

Flash

30-40 FPS

15-20 FPS

HUD
- 5.

## 1つのムービー ビュー

- Flash

C++  
Flash

( )

Flash
- 4.2
- :
1. 

Flash

Scaleform

HUD

Flash
  2. 

C++  
C++  
Render::Renderer  
C++  
AS

Flash

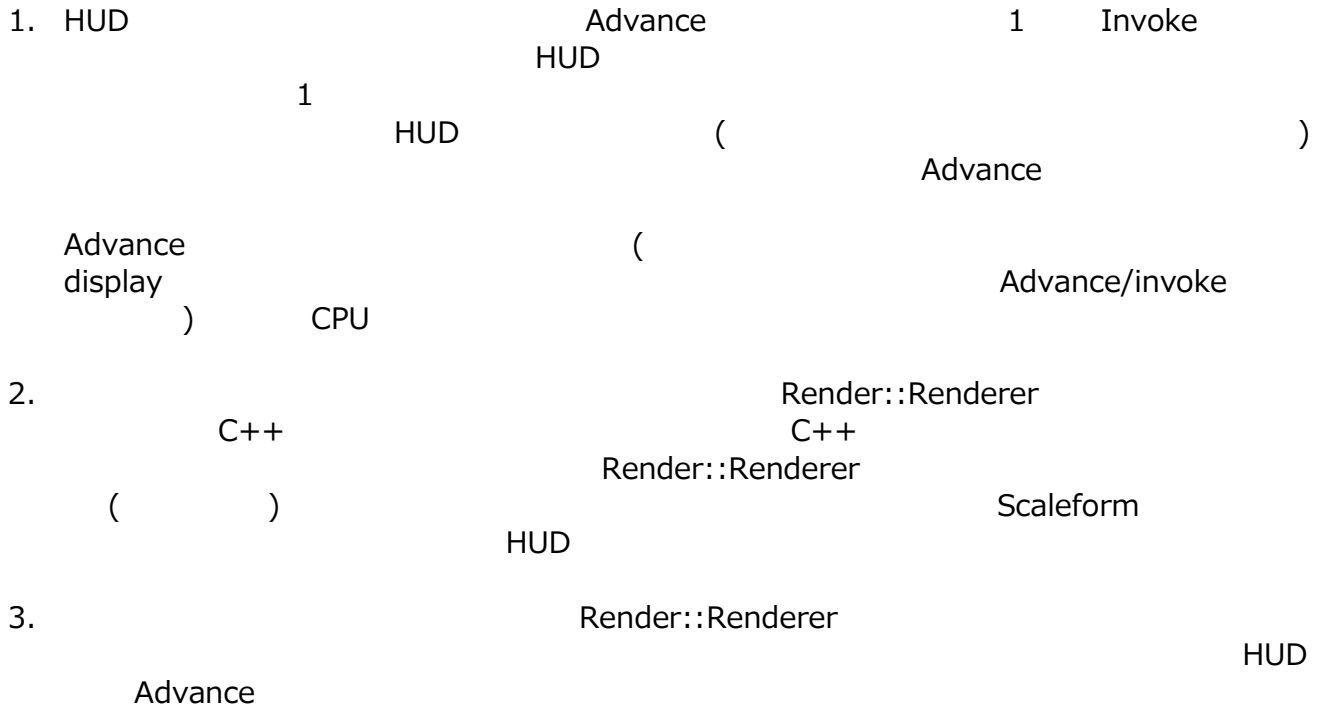
Flash  
RenderString

1

## 1つのムービー ビュー (上級編)

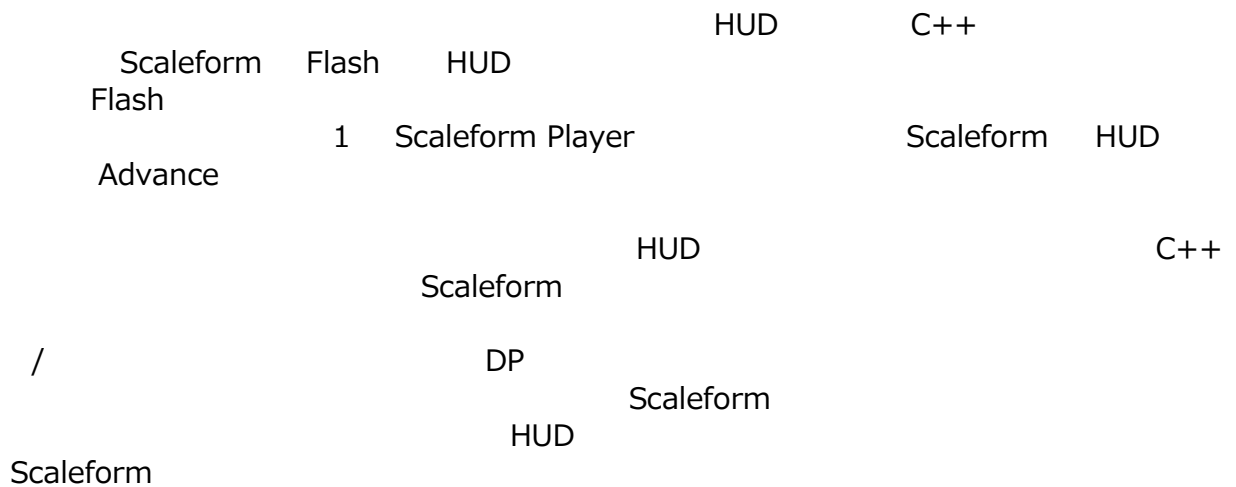
HUD  
4.3

:



: Scaleform -Unreal® Engine 3

## ***Flash* を使用しないカスタムの HUD 作成**



[Scaleform](#) / [Flash](#)  
[Scaleform](#) C++  
[DrawText API 1](#) [Scaleform 2](#) [HUD 1](#)  
[Scaleform](#) / [Scaleform FAQ](#) ["Font & Text"](#) [Font and](#)  
[Text Configuration Overview](#) / [DrawText API](#)

- HUD :
- - ["Graphics Rendering & Special Effects"](#) 1 2 FAQ
  - PC Wii™ (
    - Gfx::Movie::SetMouseCursorCount(0);
    -
  - / HUD  
noInvisibleAdvance (AS2) or scaleform.gfx.Extensions.noInvisibleAdvance (AS3)  
[\\_visible\(AS2\) or visible \(AS3\) = false](#)
  - Invoke 1 2 Invoke  
( )  
( )
  - [2.1](#)  
FAQ ["Art & Assets"](#)



## Flash のタイムライン

Flash

2

FLA

- FPS
- AS
- ([ 1] [ 2] )

## 全般的なパフォーマンスの最適化

- 
- 3 1 trace() ( AS AS actions AS 1

ActionScript 2.0 Best Practices

[http://www.adobe.com/devnet/flash/articles/as\\_bestpractices.html](http://www.adobe.com/devnet/flash/articles/as_bestpractices.html)

Flash 8 Best Practices

[http://www.adobe.com/devnet/flash/articles/flash8\\_bestpractices.html](http://www.adobe.com/devnet/flash/articles/flash8_bestpractices.html)

Flash ActionScript 2.0 Learning Guide

[http://www.adobe.com/devnet/flash/articles/actionscript\\_guide.html](http://www.adobe.com/devnet/flash/articles/actionscript_guide.html)